

Face Verification Using Siamese Network with EfficientNetBo Feature Extraction

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Introduction and Background

Face verification plays a crucial role in modern biometric systems used for security, authentication, and surveillance. Traditional methods relying on handcrafted features often fail under real-world variations like pose, lighting, and occlusion. To overcome these challenges, this project leverages a Siamese Neural Network—a deep learning architecture that compares image pairs to determine facial similarity.

By integrating EfficientNetB0 as the feature extractor, the model benefits from a lightweight yet powerful backbone that enhances accuracy while keeping computational demands low. This combination enables robust and scalable face verification, even in resource-constrained or mobile environments.

Objectives

- Design a Siamese Network using L1 distance to compare face pairs.
- Integrate EfficientNetB0 for efficient and accurate feature extraction.
- Benchmark performance against a custom CNN-based SNN.
- Ensure deployability in real-time and resource-constrained environments.

Data Construction

- Triplet-based structure:** Dataset consists of image triplets—Anchor (reference image), Positive (same person), and Negative (different person)—to train the model using similarity learning.
- Collected from public face datasets** with multiple images per identity, ensuring diverse variations in pose, lighting, and expression.
- Face detection** performed using Haar Cascade Classifier (OpenCV), ensuring cropped face regions for consistent input.
- Preprocessing:** All images are resized to 100×100 pixels and pixel values are normalized between 0 and 1 for stable model training.
- Data augmentation** applied to positive pairs to introduce realistic variations (e.g., rotation, brightness change), helping the model generalize better.
- Organized folder structure:** Images are separated into /anchor, /positive, and /negative directories for efficient data loading and training pair generation.

Methodology

- Siamese Neural Network** architecture with two identical, weight-shared branches that process input image pairs simultaneously to learn facial similarity.
- Designed to output** a similarity score rather than classify identities, making it ideal for face verification and one-shot learning tasks.

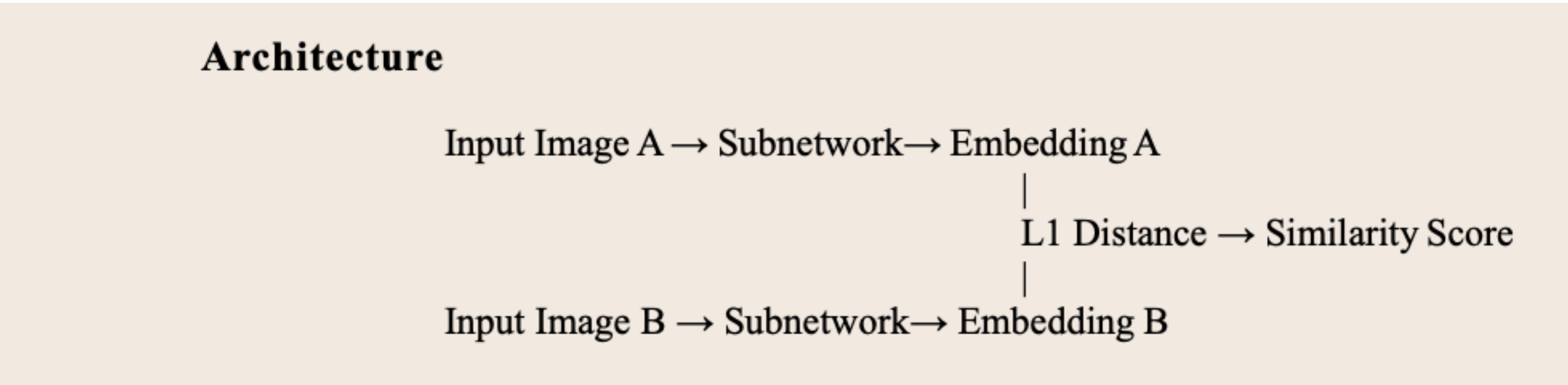
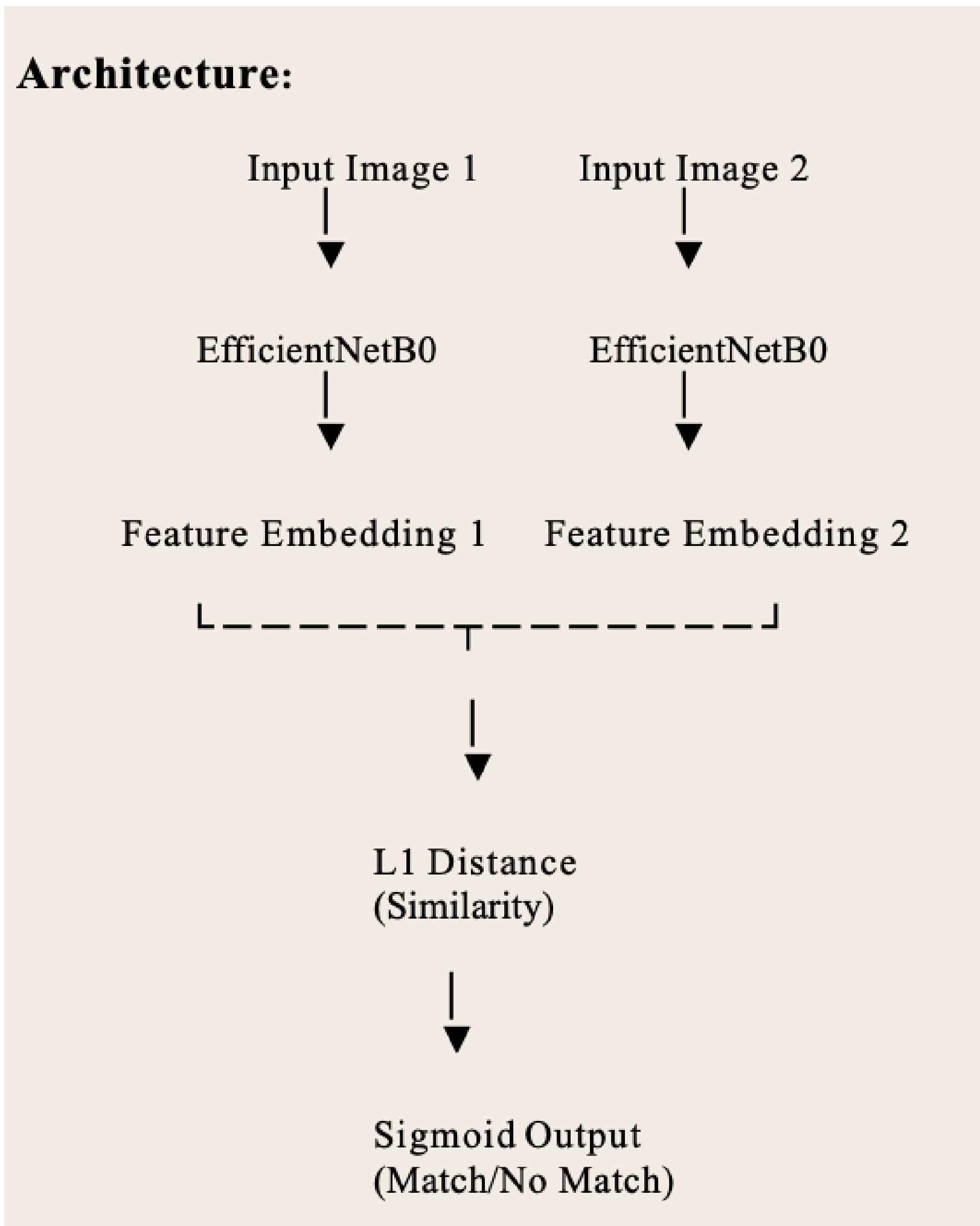


figure:Siamese Network Architecture

Integration with EfficientNetBo

- EfficientNetB0** is used as the shared backbone in both branches, replacing the custom CNN to improve accuracy and efficiency.
- Enables transfer learning from ImageNet and reduces model size, making the system scalable and suitable for real-world applications.



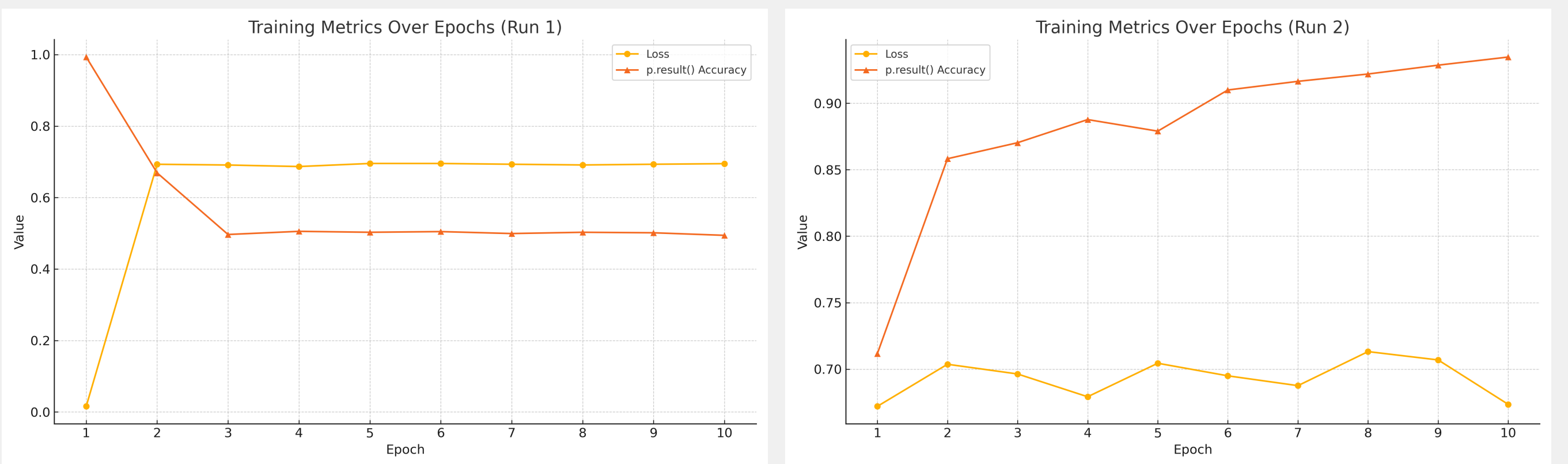
Siamese Network Architecture

- Faces are detected from images using Haar Cascade classifiers.
- Detected faces are cropped and preprocessed for model input.
- Two identical subnetworks process a pair of face images.
- Each subnetwork shares weights and extracts embeddings.
- Replaces the basic CNN backbone for better feature extraction.
- Leverages pretrained weights from ImageNet for transfer learning.
- The L1 distance is computed between the two embeddings.
- The distance vector is passed to a dense layer with a sigmoid activation, which outputs a single probability value:
Close to 1 → likely same person
Close to 0 → likely different persons
- A final sigmoid layer predicts similarity (same/different person).
- Trained on labeled image pairs with binary labels (1 for same, 0 for different).

Future Enhancements

- Integrate ArcFace loss for angular margin-based feature separation.
- Apply hard negative mining for better discrimination of similar faces.
- Use 3D face data for more robust recognition in dynamic environments.
- Optimize the model via quantization or pruning for on-device inference.

Results



Siamese Learning

Siamese with EfficientNet Learning

Metric	Baseline Siamese CNN	Siamese + EfficientNetB0
Testing Accuracy	49.39%	86.67%
Total Parameters	3,89,64,545	93,00,644
Training Epochs	10	10 (Peak at ~50)
Feature Generalization	Poor	High
Training Stability	Moderate	Stable & Faster

Results on Test Data

Key Observations

- Despite having over 75 percent fewer parameters, the EfficientNetB0-based model trained faster and was more stable during convergence.
- These results demonstrate that integrating a pre-trained, efficient architecture significantly enhances both accuracy and deployability of face verification systems.

Conclusion Implications

This project demonstrates an effective and lightweight face verification system using a Siamese Neural Network integrated with EfficientNetB0. The model achieved high accuracy (87percent) while significantly reducing the number of parameters compared to a traditional CNN-based approach. By leveraging transfer learning and L1-based similarity comparison, the system generalizes well even in challenging, real-world conditions.

The model's compact size and high performance make it suitable for deployment in real-time applications such as secure logins, attendance systems, and smart surveillance. It also lays a strong foundation for future enhancements, including the use of ArcFace loss, hard negative mining, 3D facial data integration, and optimization for edge devices.

References

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[3] Sharma, A., Gupta, P. "Siamese Neural Networks for Face Recognition." Computer Vision and Image Understanding, vol. 219, 2023.